CHAPTER 6: RESULTS AND DISCUSSION

6.1 Introduction

The Results and Discussion chapter presents the experimental observations, performance analysis, and outcomes obtained from the implementation of the IoT Smart Irrigation and Plant Monitoring System using ESP32.

The project was tested using actual hardware components including the ESP32 microcontroller, DHT11 sensor, soil moisture sensor, OLED display, relay module, and mini water pump. The objective of testing was to verify the functionality of each component and evaluate the overall system performance under different environmental conditions.

The obtained results demonstrate that the system successfully monitors environmental parameters and performs irrigation control based on soil moisture levels.

6.2 Experimental Setup

The hardware setup consisted of:

Controller

- ESP32 Development Board (38 Pin)

Sensors

- DHT11 Temperature and Humidity Sensor
- Soil Moisture Sensor

Output Devices

- OLED Display (128 × 64)

Actuator

- Relay Module
- Mini Water Pump

Power Supply

- USB Power through Computer

The components were assembled on a breadboard and tested using Arduino IDE.

6.3 Sensor Calibration Results

Before implementing automatic irrigation, the soil moisture sensor was calibrated.

Dry Soil Calibration

Sensor placed in air.

Observed Value:

```text id="2u3bwb" Raw ADC ≈ 4095

````
This value was stored as:

```cpp id="hh5x34"
AIR_VALUE = 4095
````

Wet Soil Calibration

Sensor placed in water.

Observed Value:

```text id="o7h3g4" Raw ADC ≈ 1200

````
This value was stored as:

```cpp id="57ehqr"
WATER_VALUE = 1200
````

Calibration Summary

Condition	Raw ADC Value
Air (Dry)	4095
Water (Wet)	1200

These values were used to calculate soil moisture percentage accurately.

6.4 Temperature and Humidity Monitoring Results

The DHT11 sensor successfully measured environmental conditions.

Sample Reading 1

Parameter	Value
Temperature	31.9°C
Humidity	46%

Sample Reading 2

Parameter	Value
Temperature	32.1°C
Humidity	47%

Observation

The sensor provided stable readings throughout the testing period.

The obtained values indicate:

- Moderate temperature conditions.
- Average humidity levels.
- Proper functioning of DHT11 sensor.

The sensor readings were successfully displayed on the OLED display and Serial Monitor.

6.5 Soil Moisture Monitoring Results

The soil moisture sensor successfully measured soil conditions.

Sample Reading

```text id="rr0b6g" Raw Soil Value = 1535
```

```
Calculated Moisture:

```text id="x7f5r4"
Moisture = 88%
```

Moisture Conversion Formula

```cpp id="k0x9uk" moisturePercent = map(rawValue, AIR_VALUE, WATER_VALUE, 0, 100);

```

Sample Observations

Raw Value	Moisture (%)
4095	0%
3200	30%
2500	55%
1535	88%
1200	100%

Observation

The moisture percentage increased as water content increased.

The sensor responded quickly to changes in soil conditions.

6.6 OLED Display Results

The OLED display successfully showed all required information.

Displayed Parameters:

- Temperature
- Humidity
- Soil Moisture Percentage
- Pump Status

```

````
### OLED Output Example

```text id="0nukq2"
Temp : 32.1 C

Hum  : 47 %

Soil : 88 %

Pump : OFF
````

Observation

The OLED screen provided:

- Clear display output
- Fast updates
- Real-time monitoring

The display significantly improved user interaction with the system.

6.7 Relay Module Testing Results

The relay module was tested independently before integration.

Test Procedure

GPIO26 was manually switched between HIGH and LOW states.

Observed Results:

GPIO26 State	Relay Status
LOW	Activated
HIGH	Deactivated

Observation

The relay was identified as:

```text id="gq62mz" Active LOW Relay

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This means:

LOW → Relay ON

HIGH → Relay OFF

The relay responded correctly to ESP32 control signals.

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# 6.8 Water Pump Testing Results

The mini water pump was tested separately using batteries.

### Observation

The pump successfully:

- Started when power was applied.
- Pumped water continuously.
- Operated within the specified voltage range.

The pump was suitable for small-scale irrigation applications.

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# 6.9 Automatic Irrigation Results

The automatic irrigation logic was implemented using a moisture threshold.

### Threshold Value

```cpp id="fy4b5o"
40%
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### Decision Logic

| Moisture (%) | Pump Status |
|---|---|
| Below 40% | ON |
| Above 40% | OFF |

---

### Example Test

Observed Moisture:

```text id="1vx81t" 88%

```
Expected Pump Status:

```text id="6kw5wc"
OFF
```

Serial Output:

```text id="k1d4gg" Pump: OFF

```
The software logic correctly determined that irrigation was not required.

6.10 Serial Monitor Results

The Serial Monitor displayed all system parameters successfully.

Sample Output

```text id="5p0s7t"
Temperature: 32.10 C

Humidity: 47.00 %

Raw Soil: 1535

Moisture: 88 %

Pump: OFF
```

Benefits

The Serial Monitor was useful for:

- Debugging
- Sensor verification
- Calibration

• System testing

6.11 System Performance Analysis

The overall system performance was evaluated based on several factors.

Accuracy

The sensors provided stable and repeatable readings.

Response Time

The system responded within a few seconds after environmental changes.

Reliability

The system operated continuously without crashes.

User Interface

OLED display provided clear and readable output.

Automation

The relay control logic successfully implemented automated irrigation functionality.

6.12 Advantages Observed During Testing

The following advantages were observed:

- Real-time environmental monitoring.
- Automatic irrigation capability.
- Reduced manual effort.
- Low-cost implementation.
- Easy hardware integration.
- Expandable architecture.

6.13 Challenges Faced

During implementation, several challenges were encountered.

Relay Logic Confusion

The relay module used Active LOW logic.

Initially:

- Relay behavior appeared opposite to expectations.

Solution:

- Logic was modified accordingly.

USB Port Detection Issue

Problem:

- ESP32 COM port disappeared when multiple components were connected.

Cause:

- Wiring and power distribution issues.

Solution:

- Individual component testing.
- Rechecking breadboard connections.

Pump Control Issue

Problem:

- Pump continued running despite software showing Pump OFF.

Possible Causes:

- Incorrect relay wiring.
- COM and NC terminal confusion.
- Active LOW relay behavior.

This issue highlighted the importance of hardware verification in embedded systems.

6.14 Discussion

The developed system successfully achieved its objectives.

The DHT11 sensor accurately measured environmental conditions while the soil moisture sensor monitored soil water content. The ESP32 processed sensor data efficiently and displayed information on the OLED screen.

The relay module enabled automated control of the water pump, demonstrating the practical implementation of smart irrigation principles.

The project confirms that low-cost sensors and microcontrollers can be effectively used to develop intelligent irrigation systems suitable for educational, domestic, and agricultural applications.

6.15 Summary

The experimental results demonstrate that the IoT Smart Irrigation and Plant Monitoring System operates successfully. The system accurately monitors temperature, humidity, and soil moisture while providing automatic irrigation functionality through relay-controlled pump operation.

The obtained results validate the effectiveness of the proposed system and establish a strong foundation for future enhancements such as cloud connectivity, mobile applications, and remote monitoring.